Antifungal Resistance in Dermatophytes – Review of the Epidemiology, Diagnostic Challenges and Treatment Strategies for Managing *Trichophyton indotineae* Infections

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Supplementary Online Material

Expert Review of Anti-infective Therapy

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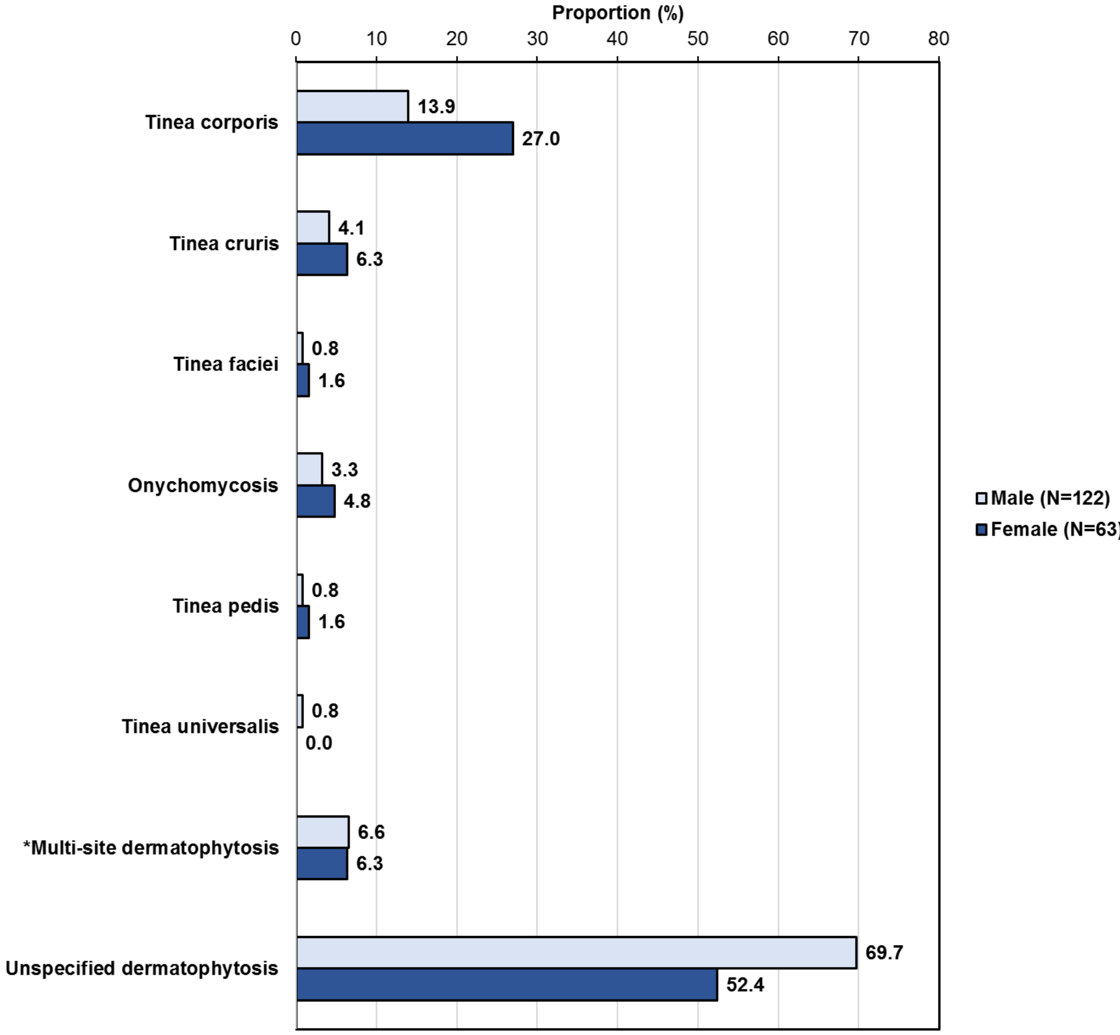

Figure S1. Summary of diagnosis pattern among male (N=122) and female patients (N=63) infected with *T. indotineae* [1–55]. *Dermatophytosis affecting multiple anatomical sites were most frequently reported as tinea corporis and tinea cruris.

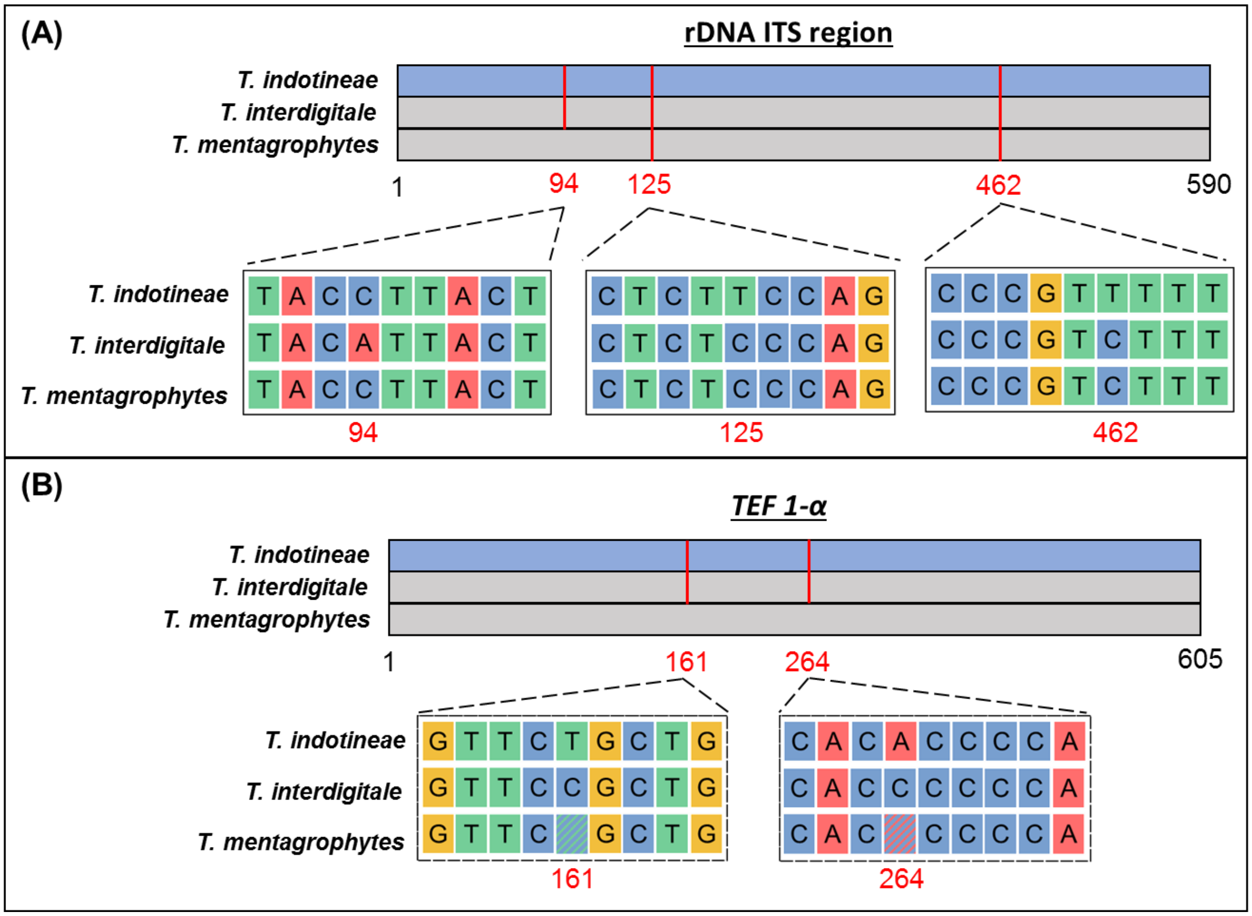

Figure S2. Nucleotide substitutions located at the (A) rDNA ITS region, and (B) the *TEF 1-α* gene, enabling differentiation of *T. indotineae* from *T. mentagrophytes* and *T. interdigitale* [30,56–58]. Note that there is only a 2 bp difference between *T. indotineae* and *T. mentagrophytes* at the ITS region; substitutions at the *TEF 1-α* gene may be found in both *T. indotineae* and selected *T. mentagrophytes* isolates.

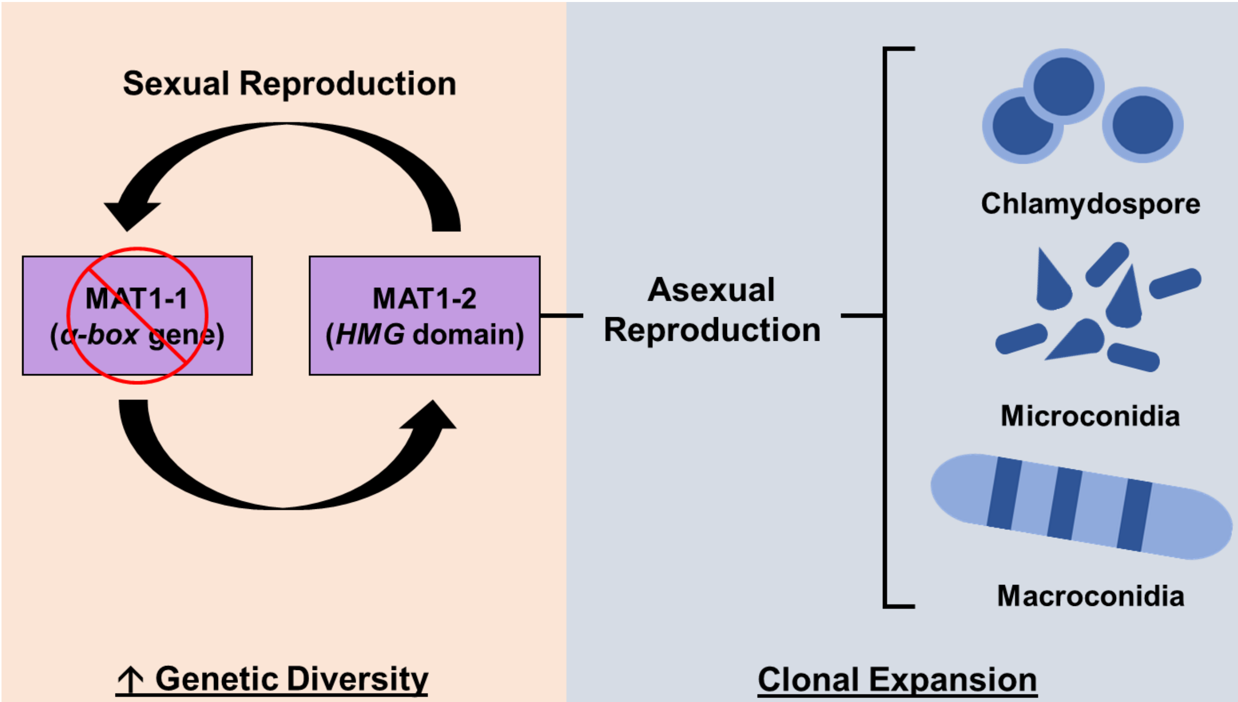

Figure S3. Mating types governing sexual and asexual reproduction in dermatophytes; each mating type (MAT1-1 or MAT1-2) is indicated by the presence of the *HMG* domain or the *α-box* gene at the *MAT* locus [59]. *T. indotineae* do not exhibit the MAT1-1 mating type and thus is limited to asexual reproduction favoring clonal expansion [30,58,60].

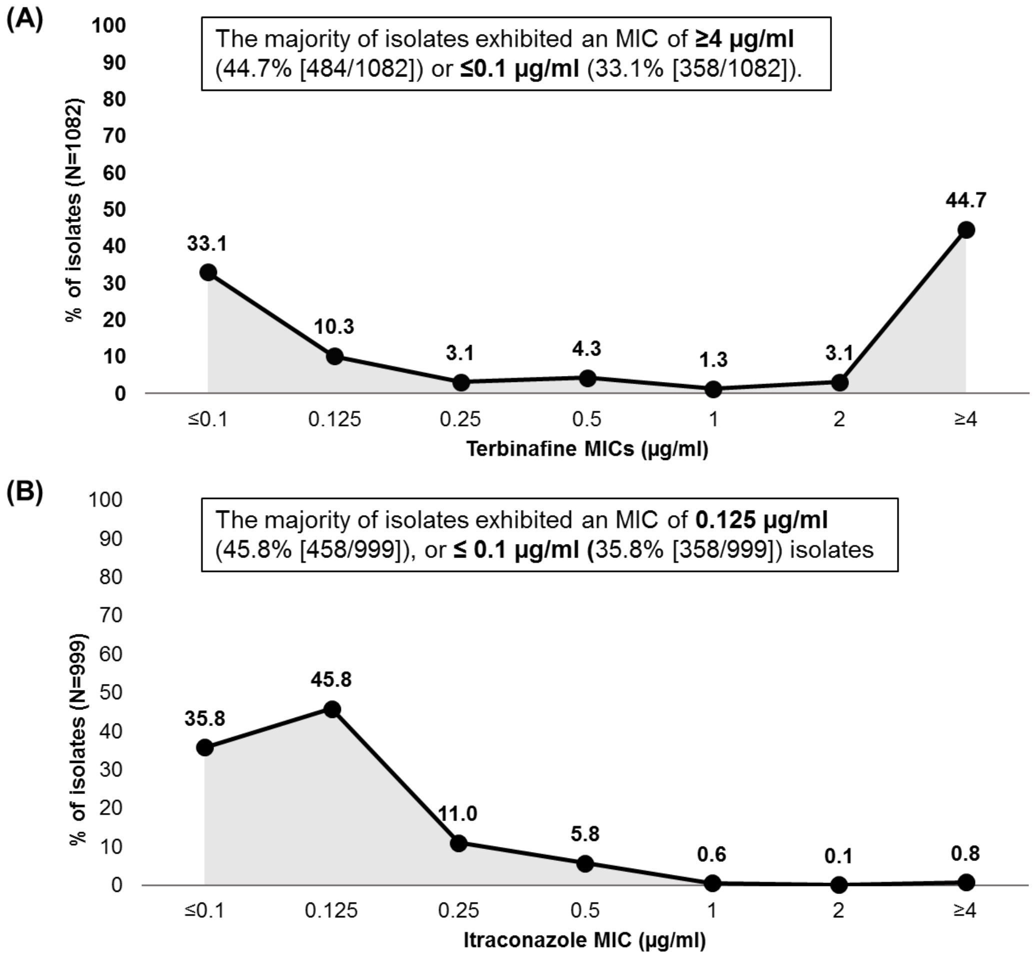

Figure S4. Distribution of (A) terbinafine MICs (N=1082) and (B) itraconazole MICs (N=999) in *T. indotineae* isolates [1–50,52–55,61,62].

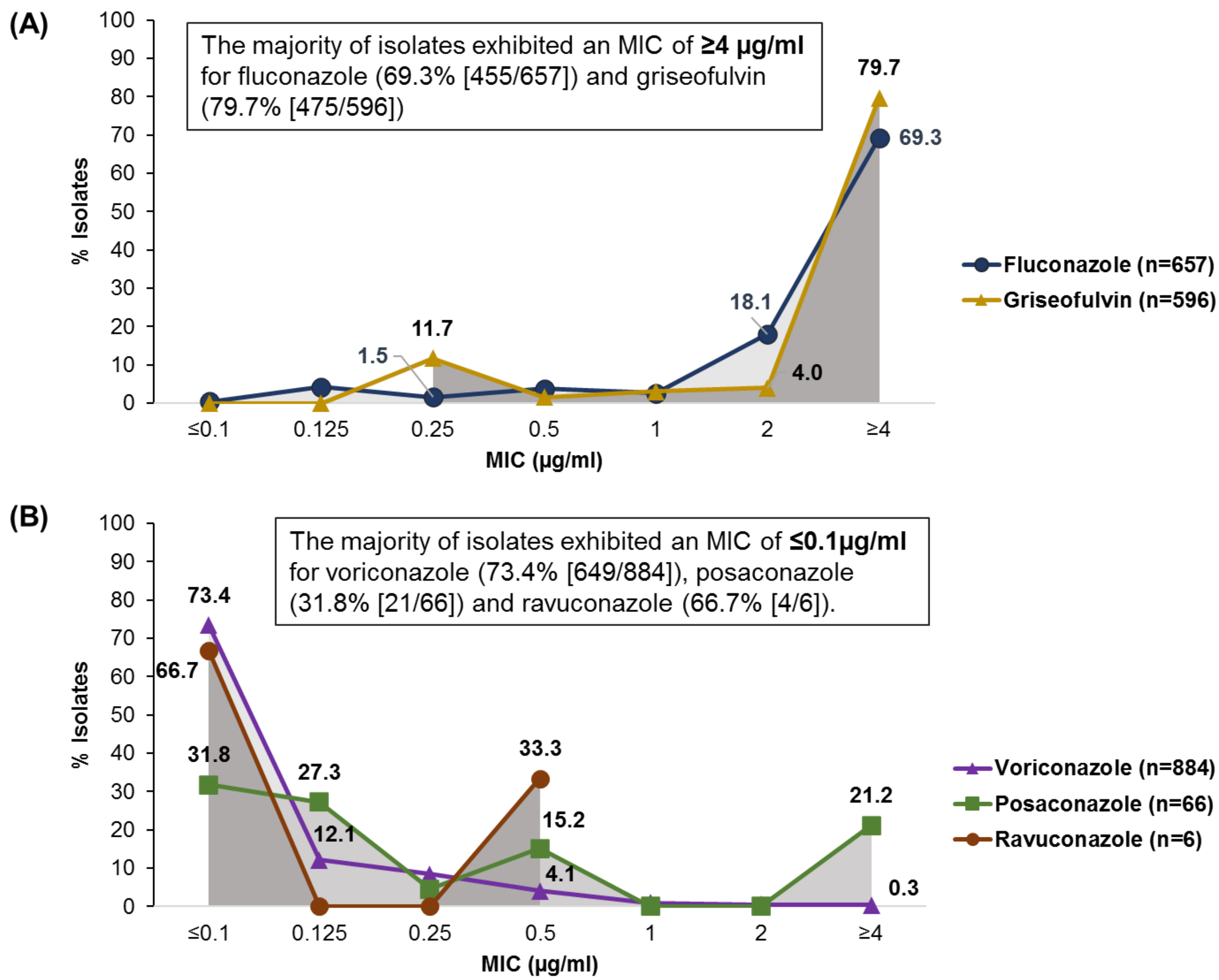

Figure S5. Distribution of (A) fluconazole and griseofulvin MICs, (B) voriconazole, posaconazole and ravuconazole MICs, in *T. indotineae* isolates [1–4,6–10,12,14–55].

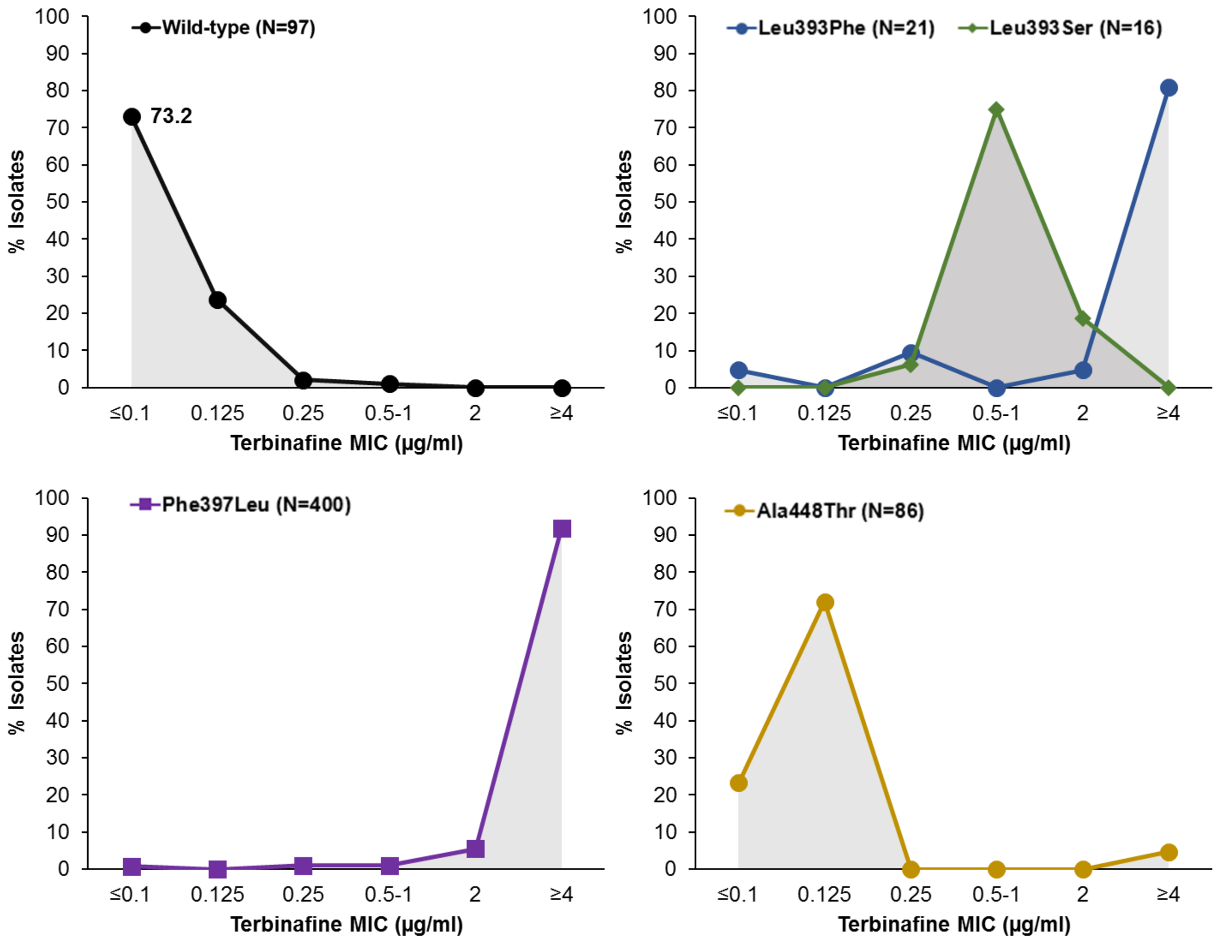

Figure S6. MIC distribution pattern of wild-type *T. indotineae* isolates (N=97), and *T. indotineae* isolates harbouring *SQLE* mutations (Leu393Phe [N=21], Leu393Ser [N=16], Phe397Leu [N=400], Ala448Thr [N=86]) [3,4,6–9,14,16–23,25–30,32–35,38–41,44–47,51,53,55].

Table S1. Surveillance data of studies included in the review

Reference	Country	Testing population	Sample	No. *T. indotineae* isolate
Abastabar 2022 [1]	Iran	Tinea capitis	Hair	5
Abdolrasouli 2024 [2]	UK	Tinea	Skin	1
Astvad 2022 [3]	Denmark	NR	NR	7
Bidaud 2022[4]	India	NR	Skin, hair, and nail	16
Bidaud 2023 [5]	France	NR	NR	8
Blanchard 2023 [6]	Switzerland	NR	Skin, nail	3
Bortoluzzi 2023 [7]	Italy	Tinea corpois	Skin	4
Brasch 2021 [8]	Germany	Tinea	Skin	5
Burmester 2022 [9]	Germany	Tinea corpois/cruris	NR	6
Burmester 2019 [10]	Germany	Tinea corpois	Skin	1
Canete-Gibas 2023 [11]	USA	NR	NR	21
Caplan 2023 JAMA [12]	USA	Tinea corpois	Skin	1
Caplan 2023 [13]	USA	Tinea corpois and onychomycosis	Skin	2
Caplan 2024 [14]	USA	NR	NR	11
Crotti 2023 [15]	Italy	Tinea corpois and onychomycosis	Skin, nail	1
Dashti 2023 [16]	Kuwait	Tinea	Skin	1
De Paepe 2024 [17]	Germany	NR	Skin	20
Delliere 2022 [18]	France	Tinea corpois	Skin	7
Durdu 2023 [19]	Turkey	Tinea cruris	Skin	2
Ebert 2020 [20]	India	Tinea	Skin	279
Fattahi 2021 [21]	Iran	Tinea, tinea pedis	Skin	4
Fukada 2024 [22]	Japan	Tinea faciei	Skin	1
Gawaz 2021 [23]	Germany	Tinea corpois	NR	1
Gueneau 2022 [24]	France	NR	Skin	1
Haghani 2023 [25]	Iran	NR	NR	40

Harada 2021 [26]	Japan	Tinea corpois	Skin	1
Hiruma 2023 [27]	Japan	Tinea corpois	Skin	2
Jabet 2022 [28]	France	Tinea corpois/cruris	NR	7
Jia 2023 [29]	China	Tinea corpois and cruris	Skin	2
Kano 2020 [30]	Japan	Tinea corpois	NR	2
Khurana 2021 [31]	India	Tinea corpois/cruris	Skin	21
Klinger 2021 [32]	Switzerland	Tinea capitis/corpois/genitalis/glutealis/pedis	Skin	21
Kolarczykova 2024 [33]	Czech Republic	Tinea unguium	NR	7
Kong 2021 [34]	China	Tinea	Skin, nails	64
Kong 2023 [35]	China	Tinea corpois/cruris	Skin	1
Kumar 2022 [36]	India	NR	Skin, nails	2
Messina 2023 [37]	Argentina	Tinea corpois	Skin	1
Moreno-Sabater 2022 [38]	France	Onychomycosis/tinea pedis/manuum/cruris/corporis	Skin, nails	6
Nenoff 2020 [39]	Germany	Tinea	Skin	29
Ngo 2022 [40]	Vietnam	Tinea	Skin	1
Pashootan 2022 [41]	Iran	Tinea	NR	10
Pavlović 2024 [42]	UAE	Dermatophytosis	Skin	1
Posso-De C 2022 [43]	Canada	Tinea	Skin	8
Russo 2023 [44]	Switzerland	Tinea corpois/cruris	Skin	2
Shaw 2020 [45]	India	NR	Skin, hair, and nail	498
Singh 2021 [46]	India	Onychomycosis, tinea corpois, cruris, capitis	Skin, nail	90
Siopi 2021 [47]	Greece	Tinea	Skin, nail	9
Spivack 2024 [48]	USA	Tinea genitalis	Skin	1
Taghipour 2020 [49]	Iran	NR	NR	28
Tamimi 2024 [50]	Iran	Tinea corpois/cruris	Skin	53
Teo 2024 [51]	Singapore	Tinea, tinea cruris	Skin	2
Thakur 2023 [52]	India	Tinea universalis	Skin	1
Uhrlaß 2024 [53]	Cambodia	Dermatophytoses	Skin	1

Villa-Gonzales 2023 [54]	Spain	Tinea corpois	Skin	1
Xie 2024 [55]	China	Tinea	Skin	14

NR, not reported

Appendix S1: References

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